

6ELEN018W - Tutorial 3 Exercises

Tutorial sessions are practical sessions that you need to work towards the exercises set. They will give you the chance to practice the material learned in the lectures and learn new things as well.

You should spend the tutorial session attempting these with the help of your tutor. Make sure that you seek the help of your tutor for anything you do not understand or whenever you are stuck with a task.

Exercise 1

Given the observable Markov model with three states S_1, S_2, S_3 , initial probabilities

$$\Pi = [0.5 \quad 0.2 \quad 0.3]^T$$

and transition probabilities

$$A = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

implement (in Python) a program which generates 100 sequences of 1000 states.

Exercise 2

Using the data generated by the previous exercise, estimate Π, A and compare with the parameters used to generate the data. A Python program should be written for such an estimation.

Exercise 3

1. Implement in Python the Markov decision process for the toymaker example seen in the lecture.
2. Extend the code by implementing a Python function accepting a single argument n . The function returns the probabilities of the toymaker to be successful and not successful after n weeks. Test your function for different large values of n .

Exercise 4

An insurance company has estimated that the conditional probability for a driver to be involved in at least one accident is 10% given that he/she was not involved in any accident this year, and it is 40% given that he/she was involved in at least one accident this year. Their statistics also show that during the very first year of driving, 80% of the drivers are not involved in any accidents. Let

$$X_n = \begin{cases} 0, & \text{if no accidents year } n \\ 1, & \text{if at least one accident year } n \end{cases}$$

for $n \geq 0$.

Assume $\{X_n \mid n = 0, 1, \dots\}$ is a Markov chain.

1. Write the transition matrix for $\{X_n \mid n = 0, 1, \dots\}$. What is the initial distribution of $\{X_n\}$, i.e. $P(X_0 = 0)$ and $P(X_0 = 1)$?
2. Calculate $P(X_3 = 0 | X_0 = 1)$.
3. Determine the probability distribution of X_1 , i.e. calculate $P(X_1 = 0)$ and $P(X_1 = 1)$.

Exercise 5

1. Implement in Python the Markov model for the recycling robot described in the lecture.
2. Run a simulation for 1000 time steps. At each step the robot takes a random action among the 3 possible actions. Experiment with different values of a and b .
3. What is the probability that the robot will be at state “High” or “Low” after 1000 time steps?

Hint: Run a large number of simulations to estimate these probabilities.